Multispectral and SAR Image Fusion for Multiscale Decomposition Based on Least Squares Optimization Rolling Guidance Filtering

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***Abstract*— Multispectral and synthetic aperture radar (SAR) image fusion is one of the key technologies to improve image quality. The fusion method of multiscale decomposition includes two aspects: the decomposition of the image and the design of the fusion rule. There are some problems in the traditional decomposition methods, such as gradient reversals, halos, and other artifacts, and limited scale separation of space overlapping features. In addition, the quality of fusion images is greatly affected by the fusion rule design. Therefore, a novel method based on least squares (LS) optimization rolling guidance filtering (RGF) for multiscale decomposition is proposed. All gradients of RGF are optimized by LS to suppress artifacts, such as gradient inversion, and then combined with Gaussian filtering for image decomposition to eliminate interference texture and speckle noise while preserving edge details. At the same time, the decomposition of the image is extended to multiscale space to achieve scale separation of space-overlapping features, which is convenient for multilevel fusion of image features. In the end, based on the scale of decomposition, the results fall into three layers, and coupled neural P (CNP) system and other rules are designed for different layers of information fusion. The results indicate that this method has a good visual effect and outperforms all the other comparison methods on nine evaluation indexes.**

***Index Terms*— Least squares (LS) optimization, multilayer image fusion, multiscale decomposition, pulse coupled P system, rolling guidance filtering (RGF).**

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I. Introduction

IMAGE fusion is designed to realize the complementary advantages and redundancy control between different sensor data of the same scene to generate a fused image with more accurate and rich information. Image fusion techniques can be categorized into three types according to the differences in imaging devices: multimodal image fusion, digital photography image fusion, and remote sensing image fusion [1]. Typical multimodal image fusion includes infrared-visible image fusion and medical image fusion [2], [3]. Digital photographic image fusion mainly involves multiexposure image fusion and multifocus image fusion [4], [5]. Remote sensing image fusion is a special kind of multimodal image fusion, which requires higher spatial preservation and spectral fidelity algorithms to enhance the resolution than multimodal image fusion [6]. Panchromatic and multispectral image fusion is the most representative scene of remote sensing image fusion. With the continuous development of remote sensing technology, the types of remote sensing images are also gradually enriched, and remote sensing image fusion also faces more and more challenges, such as the fusion of synthetic aperture radar (SAR) and multispectral images [7].

SAR is an all-day and all-weather active earth observation system. SAR image contains rich backscattered information, good spatial contrast, and can provide fine texture and structure information. However, there is serious speckle noise interference in SAR images, and its interpretability is poor [8], [9]. A multispectral sensor is a passive imaging system, which is easily affected by atmospheric environment and other factors in the imaging process. It sacrifices the spatial resolution while maintaining the spectral resolution, resulting in poor performance of image detail information. Nevertheless, its image has rich spectral information, good recognition ability of land cover, and good interpretability [10], [11]. Therefore, the quality of remote sensing images can be effectively improved by realizing information complementarity and advantage combinations between multispectral and SAR images [12].

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The fusion of heterogeneous remote sensing images (e.g., multispectral and SAR images) is a challenging task because their imaging mechanisms and spectral properties are very different. Generally, multispectral and SAR image fusion fall into three levels: pixel level, feature level, and decision level according to the different degrees of information abstraction [13], [14]. Pixel-level fusion can provide more accurate and richer detailed information, which is conducive to further interpretation and application. It includes four major categories: component substitution (CS), multiscale transform (MST), model optimization (MO), and hybrid methods [15]. Among them, CS methods are more classic and are often used in panchromatic and multispectral image fusion. For example, principal component analysis (PCA) [16], intensity-hue-saturation (IHS) [17], and Gram–Schmidt (GS) [18]. These methods can retain global details of images well, but they are sensitive to local differences between images, especially for SAR and multispectral image fusion with different luminance responses and contrast, which may result in serious spectral distortion (SD) [19]. MST methods usually divide multispectral and SAR images into high and low-frequency components, and design fusion rules correspond to their spectral signatures dual-tree complex wavelet transform (DTCWT) [20], wavelet transform (WT) [21], and multiscale geometric analysis are all classical MST methods. Multiscale geometric analysis includes curvelet transform (CVT) [22], contourlet transform (CT) [23], shearlet transform (ST) [24], non-subsampled contourlet transform (NSCT) [25], and non-subsampled shearlet transform (NSST) [26]. Comparatively speaking, multiscale geometric analysis can better show the edges and curves of an image. The fusion quality of MST methods depends largely on the fusion rule setting in addition to the decomposition structure. The MO method has also been extended to multispectral and SAR image fusion, where the strategy is to regard image fusion as a process of image restoration, establish the relationship model between the images to be fused and optimize the energy function, and obtain the fusion image by optimizing the model [27]. MO methods mainly include variational optimization models and sparse representation models, which require prior knowledge support, complex model construction, and high operation costs [28]. In addition, deep learning methods have received widespread attention in recent years due to their excellent feature extraction and image reconstruction fusion capabilities. They mainly include the image fusion frameworks of convolutional neural networks (CNNs) and generative adversarial networks, which realize feature extraction, fusion, and image reconstruction through end-to-end training and generator-discriminator adversarial games, respectively [29]. However, there are problems such as the lack of real fusion training data, the difficulty of explaining the fusion mechanism, and the overly subjective definition of the loss function of the training model [30]. The hybrid method is a combination of the previous methods, which makes comprehensive use of the advantages of all kinds of methods. Iervolino et al. [31] proposed an image fusion method combining generalized IHS transform and Atrous WT for land classification tasks. Shao et al. [17] combined IHS and gradient transfer filtering (GTF), and this method is able to retain both spatial details and radiometric information. Zhang et al. [12] conducted a fusion study of multispectral and SAR images based on Laplace's pyramid and sparse representation theory. These methods can better combine the advantages of multiple methods, for this reason, a mixed method is developed in this article. This method combines the advantages of IHS transform, multiscale decomposition, coupled neural P (CNP) system, Laplacian energy, and other methods to comprehensively improve the fusion quality of remote sensing images. This article focuses on image decomposition and fusion rule design.

Multiscale decomposition has been a wide concern in this field. In fact, it is to obtain images with different blur degrees through a series of filtering processes, and further decompose the details and background information of different scale layers. Edge-preserving filtering (EPF) has attracted wide attention from researchers because of its ability to preserve edge details while obscuring images [32]. Bilateral filtering (BF) is a classic EPF that comprehensively processes images through two kernel functions, pixel spatial domain, and pixel range domain. The pixel spatial domain kernel allocates weights based on the proximity of pixel values, while the pixel range kernel allocates weights based on the neighborhood distance of pixels [33]. Subsequently, on the basis of BF, joint BF (JBF), also known as cross BF (CBF), is derived. The stability of BF weight is improved by using a guide graph as the basis for calculating the weight of the range. Shreyamsha Kumar [34] used JBF to extract details from the original image and fused the image by weighted average. Zhang et al. [35] proposed a scale-sensing rolling guidance filtering (RGF), which performs iterative processing on JBF and can control image detail smoothing at a certain scale. Jian et al. [36] decomposed the image through RGF and optimized the image fusion according to JBF and weight graph. However, the fusion method based on RGF and other local edge-preserving filtering decomposition still has some shortcomings as follows:

1) It takes little consideration of spatial consistency and is prone to artifact phenomena such as gradient reversals and halos, which limits its ability to extract details in any space.
2) Without multiscale decomposition, scale separation of space-overlapping features cannot be realized, and some details in the original image are easy to be lost.
3) The design of fusion rules is simple, and the fusion rules are not properly designed according to the image features, which affects the quality of fusion images.

Considering the above issues, this article proposed a multiscale decomposition multispectral and SAR image fusion method on the basis of least squares (LS) optimization RGF. Some advantages are as follows:

1) *LS optimized RGF:* When there are few similar pixels in the neighborhood of a pixel, the Gaussian weighted average of RGF is unstable and prone to artifact phenomena such as gradient reversals and halos. To this end, LS is used to constrain all gradients of RGF, so that the gradient of the processed image will not be larger

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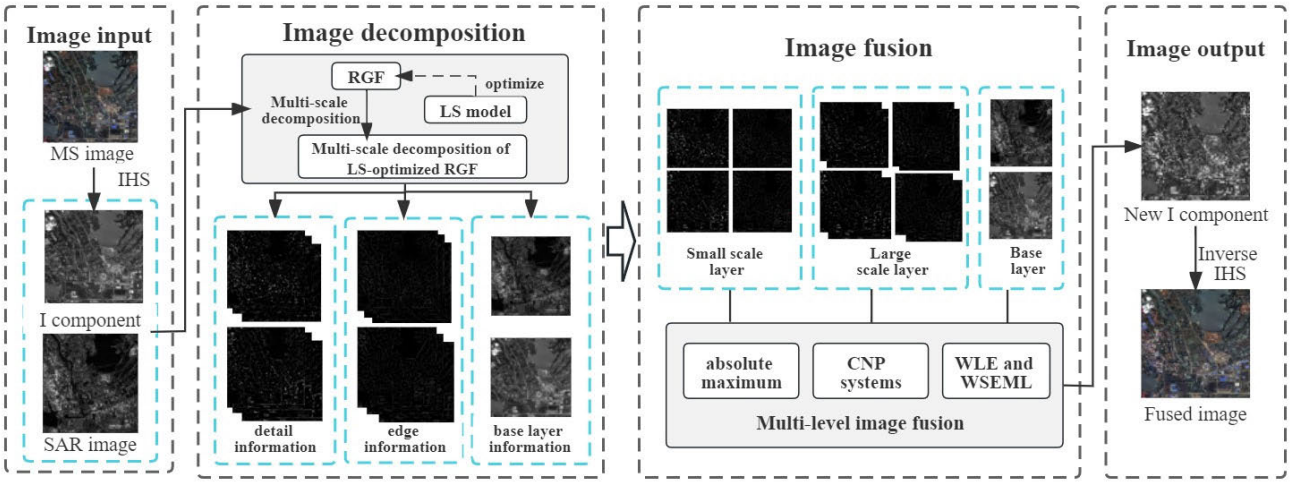


Fig. 1. Framework of the proposed fusion method.

than that of the original image, and the gradient reversals phenomenon will not occur.

2) *Multiscale image decomposition:* With the advantages of LS-optimized RGF and Gaussian filtering in noise elimination and edge preservation, the two filters are combined for image decomposition. Image decomposition is extended to multiscale space to realize scale separation of space-overlapping features. Finally, the details, edge information, and basic information are effectively separated.
3) *Multilayer image fusion:* Corresponding to the decomposition level, all the information falls into three layers, and different fusion strategies are designed accordingly. The absolute value maximization principle is adopted for the fusion of small-scale layers. CNP system is adopted for large-scale layer fusion to enhance detail information extraction. The rule of combining weighted local energy and the weighted sum of eight-neighborhood-based modified Laplacian is adopted for base layer fusion, and detailed information is extracted while the energy is preserved.

To verify the effectiveness of the method, a comparative experiment is conducted and the results are evaluated.

In Section II, multiscale decomposition of LS optimized RGF and the rule design for multilayer image fusion is described in detail. Section III introduces the experimental design, including experimental data, comparison experiments, evaluation index, and parameter setting. Section IV carries on qualitative and quantitative analysis of fusion results. Finally, Section V draws some conclusions.

II. Methodology

The methods in this article mainly include image decomposition and fusion rule design. In the image decomposition part, a multiscale image decomposition method based on least squares optimization RGF is proposed. In the part of fusion rule design, a multilayer fusion strategy is proposed, and advanced rules such as the CNP system, WLE, and WSEML are introduced. The experimental steps mainly include four parts: IHS transform, multiscale decomposition, multilayer fusion, and IHS inverse transform. The flowchart is represented in Fig. 1 and the implementation steps are indicated as follows:

1) *IHS transform:* Intensity I, hue H, and saturation S are obtained by IHS transform of multispectral image I_{MS}, respectively.
2) *Multiscale decomposition:* Least squares optimization RGF is used to conduct multiscale decomposition of intensity component I and SAR images to obtain detail information, contour information, and basic information, respectively.
3) *Multilayer image fusion:* Corresponding to the level of decomposition, three layers are classified. Among them, the absolute value maximization principle is adopted for small-scale layer fusion, the CNP system is adopted for large-scale layer fusion, and the strategy of combining WLE and WSEML is adopted for base layer fusion. The ability of detail extraction is enhanced and the energy is conserved.
4) *IHS inverse transform:* The fusion image is obtained by IHS inverse transform of the new intensity component I' and the other two components H and S after the fusion result reconstruction.

A. Least Squares Optimization Rolling Guidance Filtering for Multiscale Decomposition

1) IHS Method: IHS is a classical color transformation method, which mainly includes IHS transform and IHS inverse transform. It converts an RGB image into three independent components: intensity (I), hue (H), and saturation (S). The vast majority of the spatial information is contained in the I component, while the spectral information is separated into the H and S components. The I, H, and S components are independent of each other, which makes it easy to perform mathematical transformations. The fusion of multispectral and

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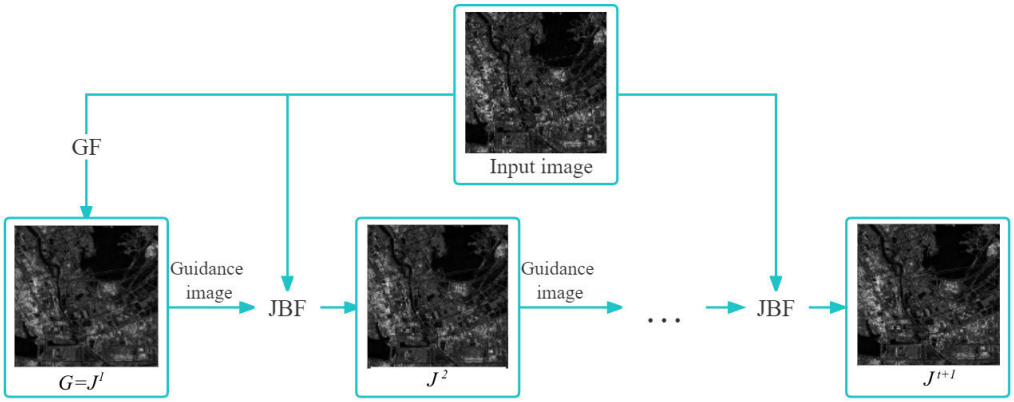


Fig. 2. Flowchart of RGF. The small image structure is first smoothed by GF and then the image edges are recovered by JBF iteration.

SAR images is usually preceded by an IHS transform of the multispectral image, and then the acquired I components are processed independently. The IHS transformation can be expressed as

$$\begin{bmatrix} I_{\mathrm{MS}} \\ v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ -\sqrt{2}/6 & -\sqrt{2}/6 & 2\sqrt{2}/6 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} \tag{1}$$

where R, G, and B denote the red, green, and blue bands of the original multispectral image, respectively, and I_{MS}, v_1, and v_2 are the I, H, and S components obtained by IHS transformation, respectively. Its inverse transformation can be expressed as

$$\begin{bmatrix} R_{new} \\ G_{new} \\ B_{new} \end{bmatrix} = \begin{bmatrix} 1 & -1/\sqrt{2} & 1/\sqrt{2} \\ 1 & -1/\sqrt{2} & -1/\sqrt{2} \\ 1 & \sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} I' \\ v_1 \\ v_2 \end{bmatrix} \tag{2}$$

where I' denote the new intensity component obtained by independent operation. R_{new}, G_{new}, and B_{new} are the new red, green, and blue bands obtained by the inverse IHS transform, respectively.

2) Rolling Guidance Filtering: RGF was first raised by Zhang et al. [35], which is a scale-sensing local edge-maintaining filter. Based on the joint bilateral filter, the RGF is realized through the iterative method of rapid convergence. It mainly consists of smoothing small image structure and iterative restoration of the image edge, and its schematic is shown in Fig. 2. Therefore, RGF is able to suppress the interference of noise in the image while enhancing the edge detail information. Gaussian Filter is a classic linear smoothing filter used to smooth out small structures in images. Its expression is as follows:

$$G_p = \frac{\sum_{q\in N(p)} \exp\left(-\frac{\|p-q\|^2}{2\sigma_r^2}\right) \cdot I_q}{\sum_{q\in N(p)} \exp\left(-\frac{\|p-q\|^2}{2\sigma_r^2}\right)} \tag{3}$$

where I_q and G_p are the input and output pixel values, respectively; p and q are the pixel indexes; and σ_r is the parameter controlling the scale of the Gaussian structure.

The edge of the GF blurred image is restored by JBF. Image G obtained through Gaussian filtering is used as the original guide graph J^1 of JBF and then iterative processing is carried out to recover edge information of different scales. The guide graph processed in each iteration is from the output image J^t of the last iteration. The expression for the procedure is

$$J_p^{t+1} = \frac{\sum_{q\in N(p)} \exp\left(-\frac{\|p-q\|^2}{2\sigma_s^2} - \frac{\|J_p^t - J_q^t\|}{2\sigma_r^2}\right) \cdot I_q}{\sum_{q\in N(p)} \exp\left(-\frac{\|p-q\|^2}{2\sigma_s^2} - \frac{\|J_p^t - J_q^t\|}{2\sigma_r^2}\right)} \tag{4}$$

where J_p^{t+1} is the result of the tth iteration, which is set as a default parameter $t = 4$ [35], [36]. σ_s is the parameter controlling the Gaussian distribution.

3) Least Squares Optimization: RGF is a scale-sensing EPF method, which refines the retained edges at a certain scale by means of rolling guidance. However, it takes little consideration of spatial consistency and is prone to artifact phenomena such as gradient reversals and halos and affects further processing of the image. The artifacts of RGF are demonstrated in Fig. 3, where white and black gradient inversion appears at the edges of the object in Fig. 3(b), and halo appears at the edges of the object in Fig. 3(e). When σ_r is small, image edges are sharpened in the filtering process and enhanced in the opposite direction, resulting in gradient reversals. When σ_r and σ_s are large, it is easy to appear halos

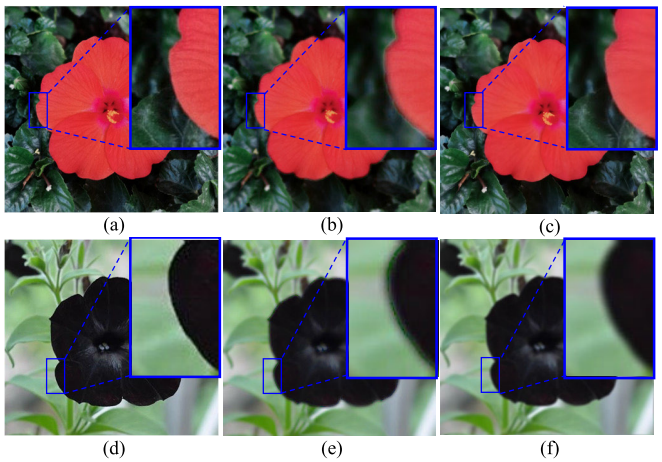


Fig. 3. Image detail comparison of different methods. (a) and (d) Input images, (b) and (e) the results of RGF, and (c) and (f) the results of LSRGF.

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phenomenon. LS optimization model consists of a data item and a regularization item, in which the data item is used to ensure the similarity between input and output images, so as to preserve edge details. The regularization term is used to control the global attributes of the output image to achieve the effect of smoothing image noise. Finally, the specific gravity between the two is balanced by an optimization solution, and smooth denoising is realized in areas with small image gradients while preserving edges in areas with large image gradients [37]. Since the regularization term of LS constrains the image gradient, it can effectively limit the generation of gradient inversion. As in Fig. 3(c) and (f), no artifacts appear as in Fig. 3(b) and (e). Therefore, LS is considered to optimize RGF, and the process can be expressed as

$$\min \sum_{i,j} \left((I_{\text{out}}(i,j) - I_{\text{in}}(i,j))^2 + \lambda \sum_{*\in(x,y)} \left(\nabla I_{\text{out},*}(i,j) - f_{\text{RGF}}\left(\nabla I_{\text{in},*}(i,j)\right)\right)\right) \tag{5}$$

where $I_{\text{in}}(i,j)$ and $I_{\text{out}}(i,j)$ denote the input and output images, respectively; λ is the parameter controlling the balance between the two; $\nabla I_{\text{out},x}(i,j)$ and $\nabla I_{\text{out},y}(i,j)$ are the directional gradients on the x and y axes of the output image, respectively; and $f_{\text{RGF}}(\nabla I_{\text{in},x}(i,j))$ and $f_{\text{RGF}}(\nabla I_{\text{in},y}(i,j))$ denote the RGF processing of the directional gradients on the x and y axes of the input image, respectively. Equation (5) can be efficiently solved by fast Fourier transform (FFT) which is expressed as

$$I_{\text{out}} = \mathcal{F}^{-1}\left(\frac{\mathcal{F}(I_{\text{in}}) + \lambda \sum_{*\in(x,y)} \overline{\mathcal{F}(\partial_*)} \cdot \mathcal{F}\left(f_{\text{RGF}}\left(\nabla I_{\text{in},x}\right)\right)}{\mathcal{F}(1) + \lambda \sum_{*\in(x,y)} \overline{\mathcal{F}(\partial_*)} \cdot \mathcal{F}(\partial_*)} \right) \tag{6}$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the FFT and its inverse transform, respectively; $\overline{\mathcal{F}(\cdot)}$ is conjugated to $\mathcal{F}(\cdot)$; ∂ denotes the delta function; and $\mathcal{F}(1)$ is the FFT of the delta function.

4) Multiscale Decomposition: In order to realize the scale separation of space-overlapping features, it is necessary to further decompose the image into extended multiscale space. Image I_r is obtained by LS-optimized RGF (LSRGF) processing of the original image I, the edge information is reserved and the details are smoothed. Compared with image I_g processed by GF, more edge information is reserved. The different methods can separate fine-scale detail D^d and edge D^e. GF and LSRGF are mixed to conduct multiscale decomposition of the original image and the construction is represented in Fig. 4. Where, the expression of detail information $D^{d,i}$, edge information $D^{e,i}$, and basic information B is

$$\begin{cases} D^{d,i} = I_g^{i-1} - I_r^i \\ D^{e,i} = I_r^i - I_g^i \\ B = I_g^N \end{cases}, \quad i = 1, 2, \ldots, N \tag{7}$$

where $I_r^i = \text{LSRGF}(I_r^{i-1}, \sigma_s, \sigma_r)$, $I_g^i = \text{GF}(I_g^{i-1}, \sigma_s)$, $I_r^0 = I_g^0 = I$, LSRGF$(\cdot)$ represents the LS-optimized RGF, GF$(\cdot)$ represents the Gaussian filtering operation, and i is the decomposition scale. The final fused image can be reconstructed by the following equation:

$$I = \sum_{i=1}^{N} \left(D^{d,i} + D^{e,i}\right) + B. \tag{8}$$

B. Multilayer Image Fusion

Multispectral images and SAR images are two kinds of images with different modes and different imaging mechanisms, so there are great differences in contrast and intensity response between them. Traditional fusion methods based on image decomposition usually decompose images into high and low frequency components, and design corresponding rules to carry out image fusion. These fusion rules are usually designed according to simple measures, which make it easy to lose some details in the original image and difficult to obtain a satisfactory fusion effect. Therefore, corresponding to the distribution of feature information, the decomposed images fall into three layers. Small-scale layer including $D_{\text{MS}}^{d,1}$, $D_{\text{MS}}^{e,1}$, $D_{\text{SAR}}^{d,1}$, and $D_{\text{SAR}}^{e,1}$, which is the first-order decomposition result of multiscale decomposition, large-scale layer including $D_{\text{MS}}^{d,n}$, $D_{\text{MS}}^{e,n}$, $D_{\text{SAR}}^{d,n}$, and $D_{\text{SAR}}^{e,n}$ $(n = 2, 3, \ldots, N)$, which is the second to the Nth order decomposition result. The base layer includes B_{MS} and B_{SAR}, which is the last-order decomposition result. Since the three scale layers focus on different image features, it is necessary to design corresponding fusion strategies for different scale layers, so as to better combine complementary information of multispectral and SAR images.

1) Small-Scale Decomposition: The strategy of absolute value maximization is used to fuse small-scale layers to reserve more texture information in the original image. It can be expressed as

$$D_F^{\psi,1} = \text{GF}\left(W_j, \sigma_0\right) D_{\text{MS}}^{\psi,1} + \left(1 - \text{GF}\left(W_j, \sigma_0\right)\right) D_{\text{SAR}}^{\psi,1} \tag{9}$$

where $\text{GF}(W_j, \sigma_0)$ denotes Gaussian smoothing of the weighting coefficients W_j to reduce the interference of noise and σ_0 is usually set to 2 and

$$W_j = \begin{cases} 1, & \left|D_{\text{MS}}^{\psi,1}\right| > \left|D_{\text{SAR}}^{\psi,1}\right| \\ 0, & \text{otherwise} \end{cases}, \quad \psi \in (d, e).$$

2) Large-Scale Decomposition: CNP system is used to fuse the large-scale layer to better utilize edge structure information and detail information. CNP system is a distributed parallel computing model that combines neuronal emergent and coupling mechanisms and has proved to be Turing's universal digital generation/reception device and functional computing device [38]. CNP system of size Π can be expressed as

$$\Pi = (O, \sigma_{11}, \sigma_{12}, \ldots, \sigma_{1n}, \ldots, \sigma_{m1}, \sigma_{m2}, \ldots, \sigma_{mn}, \text{syn}) \tag{10}$$

where $O = \{a\}$ is a single-letter alphabet (the object a is called a spike). syn $= \{(ij, kl) | 1 \le i \le m,\ 1 \le j \le n,\ |k - i| \le r, |l - j| \le r, i \ne k, j \ne l\}$ denotes a synapse when the neighborhood radius is r. $\sigma_{11}, \sigma_{12}, \ldots, \sigma_{mn}$ are an array of $m \times n$ coupled neurons, each neuron σ_{ij} having

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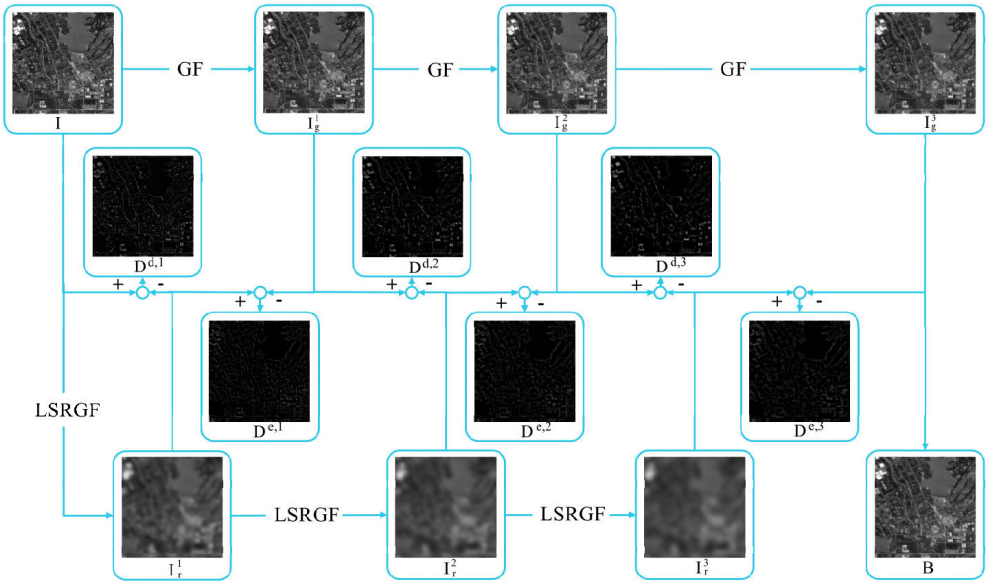


Fig. 4. Construction of a three-level multiscale decomposition.

three data units: feeding input unit u_{ij}, linking input unit v_{ij}, and dynamic threshold unit τ_{ij}, and a spiking rule set R_{ij}, i.e., $\sigma_{ij} = (u_{ij}, v_{ij}, \tau_{ij}, R_{ij})$, $1 \le i \le m, 1 \le j \le n$. where $u_{ij}, v_{ij}, \tau_{ij} \in \mathrm{R}$, R_{ij} is of the form $E/(a^u, a^v, a^\tau) \to a^p$, and E is emission condition, $0 < p < v(u+1)$, and $\tau > 0$.

Each neuron σ_{ij} has two forms of input: one is the spikes $p_{kl}(t)$ received from a neighboring neuron $\{\sigma_{kl} || k-i| \le r, |l-j| \le r\}$; the other is an external stimulus C_{ij} at image location (i, j). When the firing conditions $E \equiv (n_{ij}(t) \ge \tau_{ij}(t)) \wedge (u_{ij}(t) \ge u) \wedge (v_{ij}(t) \ge v) \wedge (\tau_{ij}(t) \ge \tau)$ and $n_{ij}(t) = u_{ij}(t)(1 + v_{ij}(t))$, neuron σ_{ij} excites, i.e., it consumes spikes with values u, v, and τ in the feeding input unit, linking input unit, and dynamic threshold unit, respectively. The equation of state for this process can be expressed as

$$u_{ij}(t+1) = \begin{cases} u_{ij}(t) - u + C_{ij} + \sum_{\sigma_{kl} \in \delta_r} w_{kl} p_{kl}(t), & \text{if } \sigma_{ij} \text{ fires} \\ u_{ij}(t) + C_{ij} + \sum_{\sigma_{kl} \in \delta_r} w_{kl} p_{kl}(t), & \text{otherwise} \end{cases} \tag{11}$$

$$v_{ij}(t+1) = \begin{cases} v_{ij}(t) - v + \sum_{\sigma_{kl} \in \delta_r} w_{kl} p_{kl}(t), & \text{if } \sigma_{ij} \text{ fires} \\ v_{ij}(t) + \sum_{\sigma_{kl} \in \delta_r} w_{kl} p_{kl}(t), & \text{otherwise} \end{cases} \tag{12}$$

$$\tau_{ij}(t+1) = \begin{cases} \tau_{ij}(t) - \tau + p, & \text{if } \sigma_{ij} \text{ fires} \\ \tau_{ij}(t), & \text{otherwise} \end{cases} \tag{13}$$

where w_{kl} is the corresponding weighting factor and p is the spikes generated when σ_{ij} is fired.

The large-scale layers of multispectral and SAR images are considered as external stimuli of Π_{MS} and Π_{SAR}, respectively, and the system is stopped when the number of iterations reaches $t_{\max}$ by setting $u_{ij}(0) = v_{ij}(0) = 0$, $\tau_{ij}(0) = \tau_0$, $u = u_{ij}(t)$, $v = v_{ij}(t)$, and $\tau = \tau_{ij}(t)$. Then, the fusion rules of the large-scale layers are defined according to the number of firings t_{ij}^{MS} and t_{ij}^{SAR} for neuron σ_{ij} in Π_{MS} and Π_{SAR}, respectively, and the expressions are

$$D_F^{\psi,n} = \begin{cases} D_{\mathrm{SAR}}^{\psi,n}, & \text{if } t_{ij}^{\mathrm{SAR}} \ge t_{ij}^{\mathrm{MS}} \\ D_{\mathrm{MS}}^{\psi,n}, & \text{otherwise} \end{cases}, \quad n = 2, 3, \ldots, N. \tag{14}$$

3) Base Layer Fusion: The base layer reflects coarse scale structure information and the overall appearance of the image. The ideal base layer does not contain detail and edge information and can be approached by setting up a lot of iterations, but this also carries a huge computing cost. The current common fusion methods of the base layer will lose the residual details and structural information after fusion because they are created for the ideal base layer. Therefore, WLE and WSEML are adopted to fully fuse the base layer B_{SAR} and B_{MS} to extract detail and edge information, where WLE mainly measures image structure information and WSEML mainly measures image detail information. The expression of WLE is

$$\mathrm{WLE}_S(i, j) = \sum_{m=-r}^{r} \sum_{n=-r}^{r} W \times (m + r + 1, n + r + 1) \times B_S(i + m, j + n)^2 \tag{15}$$

where $S \in \{\mathrm{SAR}, \mathrm{MS}\}$ and $B(i, j)$ represents the base layer at position (i, j). W is the weighting matrix of $(2r+1) \times (2r+1)$, where the value of each element of W is set to 2^{2r-d}, r is the radius of the matrix W, and d is the four-neighborhood distance from the corresponding element to the center of the matrix. For instance, when r is set to 1, the normalization matrix W can be expressed as $(1/16)\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$.

Eight-neighborhood modified Laplacian (EML) considers the effect of diagonal coefficients and can make full use of neighborhood information [39]. WSEML is a weighted

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representation of it. The expression is

$$\begin{aligned}&\mathrm{WSEML}_S(i,j)\\&\quad=\sum_{m=-r}^{r}\sum_{n=-r}^{r}W\times(m+r+1,n+r+1)\\&\qquad\times\mathrm{EML}_S(i+m,j+n)\end{aligned}\tag{16}$$

$$\begin{aligned}&\mathrm{EML}(i,j)\\&\quad=|2B(i,j)-B(i-1,j)-B(i+1,j)|\\&\qquad+|2B(i,j)-B(i,j-1)-B(i,j+1)|\\&\qquad+\frac{1}{\sqrt{2}}|2B(i,j)-B(i-1,j-1)-B(i+1,j+1)|\\&\qquad+\frac{1}{\sqrt{2}}|2B(i,j)-B(i-1,j+1)-B(i+1,j-1)|.\end{aligned}\tag{17}$$

Accordingly, the fused base layer $B_F(i,j)$ is obtained, and its expression is

$$B_F(i,j)=\begin{cases}B_{\mathrm{SAR}}(i,j), & \text{if } \mathrm{WLE}_{\mathrm{SAR}}(i,j)\cdot\mathrm{WSEML}_{\mathrm{SAR}}(i,j)\\ & \geq \mathrm{WLE}_{\mathrm{MS}}(i,j)\cdot\mathrm{WSEML}_{\mathrm{MS}}(i,j)\\ B_{\mathrm{MS}}(i,j), & \text{otherwise.}\end{cases}\tag{18}$$

III. Experiments

A. Experimental Design

1) Experimental Data: Three sets of multispectral and SAR images are selected as experimental data, all of which are from the WHU-OPT-SAR dataset (https://github.com/AmberHen/WHU-OPT-SAR-dataset). The multispectral images are from the GF-1 satellite with a spatial resolution resampled to 5 m. The SAR images are GF-3 Fine Strip II images with a sampling resolution of 5 m, as shown in Fig. 5. The GF-1 data include four bands, RGB and NIR, and the experiments are mainly focused on the three visible bands of RGB. After preprocessing such as noise reduction and registration, these two images have the same spatial position and pixel size, i.e., 1024 × 1024.

The ground truth reference source used to verify the accuracy of land cover classification is the area of interest selected based on high-resolution images, and the authenticity of individual feature class reference sources can be obtained. Due to the advantages of high spatial resolution of the experimental data in this article, the samples of interest areas are mainly selected on the multispectral images of the experimental data, and supplemented by Google Maps. A total of 100 samples from each of the five land cover types are selected and evenly distributed throughout the study area, in which 50 samples from each land cover type are randomly selected as training samples and the other 50 as validation samples.

2) Compared Methods: To better illustrate the experimental results, the designed method is compared with 16 methods, including three classical MST fusion methods, namely Curvelet [40], DTCWT [40], and NSCT [40]; three EPF fusion methods based on a single scale, namely JBF [34], RGF [36], and GTF [41]; five multiscale decomposition fusion methods, namely multiscale Gaussian BF (MSGBF) [42], multiscale Gaussian RGF (MSGRGF) [43], multiscale Gaussian curvature filtering (MSGCF) [44], multiscale decomposition methods based on significant detection and Gaussian filtering (MDSDGF) [45], and multiscale decomposition methods based on latent low rank representation (MDLatLRR) [46]. And five other methods, i.e., convolutional sparse-based morphological component analysis (CSMCA) [47], CNN [48], a method that combines IHS and GTF (IHS-GTF) [17], a method that combines Laplace pyramid and sparse representation (LP-SR) [12], and a method that combined NSST and pulse-coupled neural networks optimized by the multiscaled morphological gradient (MS-NSST-PCNN) [49]. The data for all methods came from the same experimental environment, and the parameters in these methods are derived from the default values of their respective authors.

TABLE I
Quantitative Evaluation Indexes and Their Details

Indexes	Abbreviation	Attention information	Trend (↑ or ↓)
Information Entropy	IE	Information quantity	The bigger, the better (↑)
Mutual Information	MI		
Average Gradient	AG	Spatial information	The bigger, the better (↑)
Spatial Frequency	SF		
Spectral Distortion	SD	Spectral information	the smaller the better (↓)
Relative Dimensionless Global Error in Synthesis	ERGAS		
Spectral Angle Mapper	SAM		
Quality with No Reference	QNR	Degree of distortion	The bigger, the better (↑)
Peak Signal to Noise Ratio	PSNR		

3) Evaluation Indexes: Qualitative evaluation of remote sensing images mainly relies on the subjective analysis of the whole image by the observer. By comparing fusion images with reference images, the differences in spatial details and spectral information between the two images are observed. However, the small difference between the two images is not easy to be captured by the visual system, so it needs to be further analyzed by quantitative evaluation. Quantitative evaluation is to objectively evaluate the information amount, spatial quality, spectral quality, and distortion degree of experimental results through evaluation indexes. The following nine commonly used evaluation indexes are selected in this article, as shown in Table I:

1) Information entropy (IE) is an evaluation metric to measure the amount of information contained in the fused image. Its expression is

$$\mathrm{IE}=-\sum_{i=0}^{L-1}Z_i\log(Z_i)\tag{19}$$

where L is the gray level of the image and Z_i is the statistical probability of the gray level histogram.

2) Mutual information (MI) measures the similarity between two distributions, i.e., the extent to which the fused image acquires information from the original

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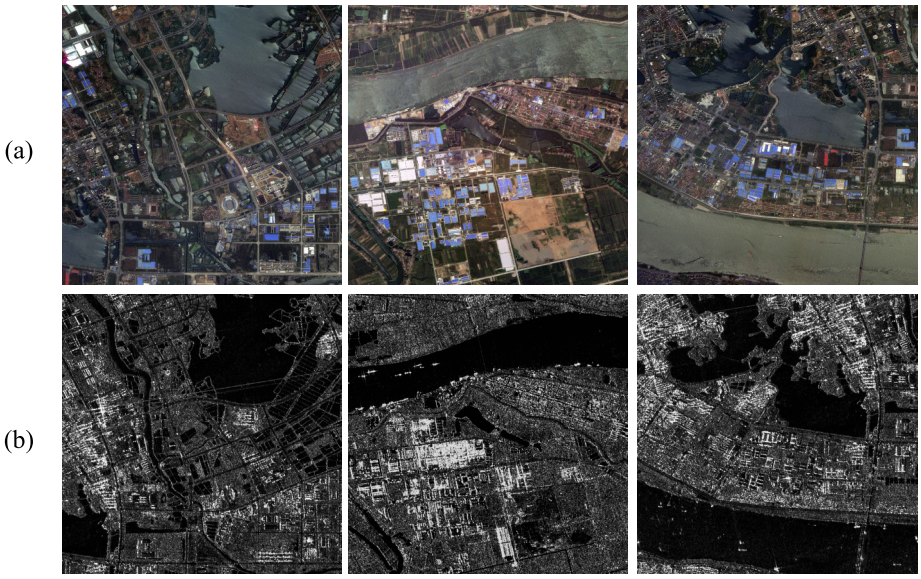


Fig. 5. Three sets of multispectral and SAR images used for the fusion experiment. (a) Multispectral image. (b) SAR image.

image. It can be expressed as

$$\mathrm{MI} = \sum_{i=1}^{M}\sum_{j=1}^{N} h_{I_A I_B}(i,j) \times \log_2\left(\frac{h_{I_A I_B}(i,j)}{h_{I_A}(i,j)h_{I_B}(i,j)}\right) \tag{20}$$

where I_A and I_B are the fused image and the reference image, respectively, and M and N are the length and width of the image, respectively. $h_{I_A I_B}$ is the joint grayscale histogram of I_A and I_B.

3) Average gradient (AG) can be used to measure the clarity of the image, the larger the AG the higher the clarity. It can be calculated as

$$\mathrm{AG} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}\sqrt{\frac{\nabla F_x^2(i,j) + \nabla F_y^2(i,j)}{2}} \tag{21}$$

where $\nabla F_x^2(i,j)$ and $\nabla F_y^2(i,j)$ denote the first-order difference of the image F in the x- and y-directions, respectively.

4) Spatial frequency (SF) can reflect the rate of grayscale change to measure the overall activity of an image. It can be expressed as

$$\mathrm{SF} = \sqrt{\mathrm{RF}^2 + \mathrm{CF}^2} \tag{22}$$

$$\mathrm{RF} = \sqrt{\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(F(i,j) - F(i,j-1))^2} \tag{23}$$

$$\mathrm{CF} = \sqrt{\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(F(i,j) - F(i-1,j))^2} \tag{24}$$

where RF and CF denote the row and column spatial frequencies of the image F, respectively.

5) SD measures the spectral difference between the fused image and the reference image. Its expression is

$$SD = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(I_A(i,j) - I_B(i,j)). \tag{25}$$

6) Relative dimensionless global error in synthesis (ERGAS) reflects the degree of SD between the fused image and the reference image. It can be calculated as

$$\mathrm{ERGAS} = 100\frac{h}{l}\sqrt{\frac{1}{K}\sum_{k=1}^{K}\frac{\mathrm{RMSE}(I_A, I_B)}{\mu_{I_A}}} \tag{26}$$

$$\mathrm{RMSE}(I_A, I_B) = \frac{1}{MN}\sqrt{\sum_{i=1}^{M}\sum_{j=1}^{N}(I_A(i,j) - I_B(i,j))^2} \tag{27}$$

where h/l is the ratio of the resolution of the SAR and MS images, K denotes the number of bands, and μ denotes the mean value of the image.

7) Spectral angle mapper (SAM) measures the similarity between the spectra by calculating the included angle between two vectors. It can be written as

$$\mathrm{SAM}(v, \hat{v}) = \arccos\left(\frac{\langle v, \hat{v}\rangle}{\|v\|_2 \cdot \|\hat{v}\|_2}\right) \tag{28}$$

where v is the spectral pixel vector of the original image and $\hat{v}$ is the spectral pixel vector of the fused image.

8) Quality with no reference (QNR) is calculated from the spatial distortion index and the SD index of the fused and source images. The complement of these two indices is multiplied to get the total QNR index. It measures the spectral and spatial distortion of the fused image. It can be expressed as

$$\mathrm{QNR} = (1 - D_\lambda)^\alpha (1 - D_\delta)^\beta \tag{29}$$

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where D_λ and D_δ represent spectral and spatial distortion, respectively. α and β are the weighting coefficients.

9) Peak signal-to-noise ratio (PSNR) measures the ratio of effective information to noise in a fused image to reflect the distortion of the fused image. Its expression is

$$\mathrm{PSNR} = 20\log_{10}\left(\frac{Z^2}{\mathrm{MSE}}\right) \tag{30}$$

$$\mathrm{MSE} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(I_A(i,j) - I_B(i,j)) \tag{31}$$

where Z denotes the difference between the maximum and minimum values of the grayscale of the ideal reference image, usually $Z = 255$ when the image is 8-bit binary.

4) Parameter Setting: The parameters of our method are set as follows. Similar to most MSD-based fusion methods, the number of multiscale decomposition levels is usually set to $N = 4$ [42], [43]. From (10)–(13), there are five priori parameters of the CNP system, namely, iterations number $t_{\max}$, initial threshold τ_0 for each neuron, value of generated spikes p in spiking rule for each neuron, neighborhood radius r, and the local weight matrix $W_{r\times r}$. In this article, we set the values of the above five parameters as $t_{\max} = 110$, $\tau_0 = 0.3$, $p = 1$, $r = 7$, $W_{7\times7} = \{w_{a,b}/w_{a,b} = (1/(\sqrt{(a-4)^2+(b-4)^2}))$, $a \neq 4, b \neq 4, 1 \leq a \leq 7, 1 \leq b \leq 7\}$ [50], [51].

To verify the sensitivity of the number of scales N of the multiscale decomposition, the number of iterations t of the RGF, and the number of iterations $t_{\max}$ of the CNP system to the experimental results, the three parameters are verified in a control variable manner and evaluated by IE, respectively. The experimental settings are as follows, when N is set to 1, 2, 3, 4, 5, 6, respectively, t is 4 and $t_{\max}$ is 110. When t is set to 1, 2, 3, 4, 5, 6, respectively, N is 4 and $t_{\max}$ is 110. When $t_{\max}$ is set to 50, 70, 90, 110, 130, 150, respectively, N is 4 and t is 4. The results of the experiments are shown in Fig. 6, with the three N, t, and $t_{\max}$ parameters leveling off around 4, 4, and 110, respectively, and then starting to increase slowly. These three parameters have a common property that the efficiency of the method decreases as the number of iterations increases. To keep the experiment as efficient as possible, the turning point that tends to level off is chosen as the best parameter setting, so the settings of the three parameters N, t, and $t_{\max}$ are reasonable.

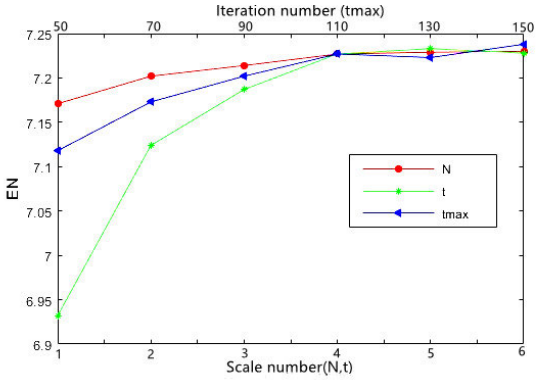


Fig. 6. Sensitivity analysis plot for parameters N, t, and $t_{\max}$.

B. Results and Analysis

The results of fusion and land cover classification are evaluated and analyzed qualitatively and quantitatively.

1) Qualitative Evaluation: Figs. 7–9 show the results of Area 1, Area 2, and Area 3, respectively, where (a)–his correspond to the original multispectral image, the fusion image of the 16 comparison methods, and the fusion image of the method in this article, in that order. As shown in Fig. 7, three MST methods do not perform well in terms of space preservation and spectral fidelity. Among them, the spatial ambiguity of NSCT is obvious, and the ability to extract details is poor. The sharpness and spectral fidelity of three single-scale EPF methods are not ideal. Because these methods cannot retain the spatial texture details and contour structure information well. As for spectral fidelity, these methods have trouble in extracting detail information in scale space, some space overlapping features are lost, making serious SD except JBF. The five multiscale decomposition methods showed varying degrees of optical distortion, and the playgrounds in the zoomed-in region showed the most pronounced distortion for MDLatLRR. The five multiscale decomposition methods have higher clarity, which is closely related to the multiscale decomposition of this kind of methods. Through the extraction of multiscale spatial details, more texture details and structure contour information are captured. However, this kind of method has some shortcomings in the performance of spectral fidelity, the SD of MSGBF and MDLatLRR is the most serious, and the color of ground objects such as buildings, bare ground, vegetation, and water bodies has obvious distortion. Among the five other methods, LP-SR has better clarity, MS-NSST-PCNN has better spectral fidelity, and CSMCA, CNN, and IHS-RGF have average spatial and spectral quality with no significant advantage. Compared with the 16 comparison methods, the method in this article clearly shows detail information of the edges and contours of buildings, roads and water bodies, and better represents characteristics of object types in different places, showing its advantages in preserving spatial details. Its fusion image is highly consistent with the original multispectral image, reflecting its excellent spectral fidelity.

According to Fig. 8, the spatial clarity of three MST methods is not significantly improved, there is a certain SD phenomenon, and the color of water bodies and bare ground is darker. Three single-scale EPF methods have poor performance in spatial preserving and spectral fidelity. In terms of spatial details, the three methods are fuzzy in the performance of the edge and contour information of ground objects, such as roads and buildings, especially the spatial ambiguity of GTF. In terms of spectral fidelity, spectral aberrations are observed for each place, such as the buildings in the zoomed-in region were obviously darkened. Among the five multiscale decomposition methods, MSGRGF and MSGCF have better spatial clarity and spectral fidelity, clear details of ground objects, and no obvious SD. However, the SD of MSGBF and MDLatLRR is obvious, which cannot accurately show the real features of ground objects. The spectral fidelity of

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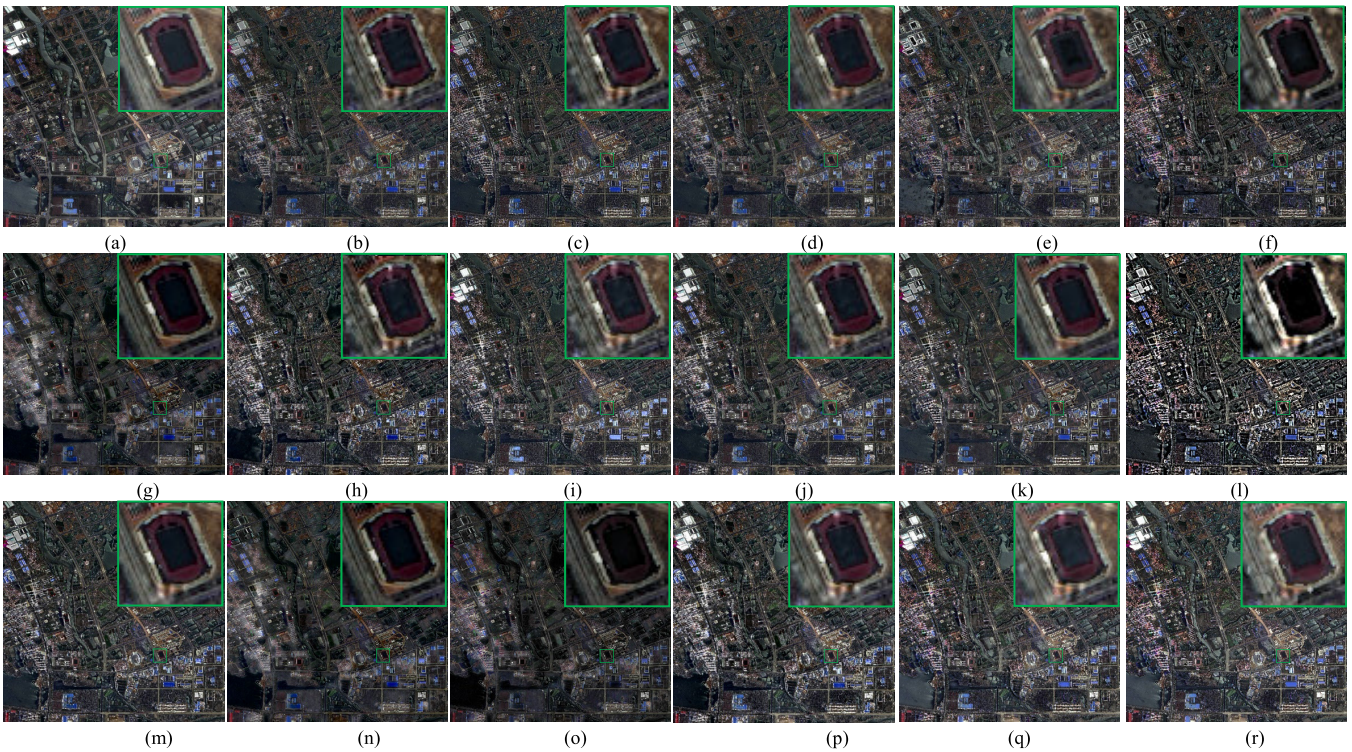


Fig. 7. Fusion results of different methods in area 1. (a) Original multispectral image, (b)–(q) fusion images of the 16 compared methods, and (r) fusion image of the proposed method.

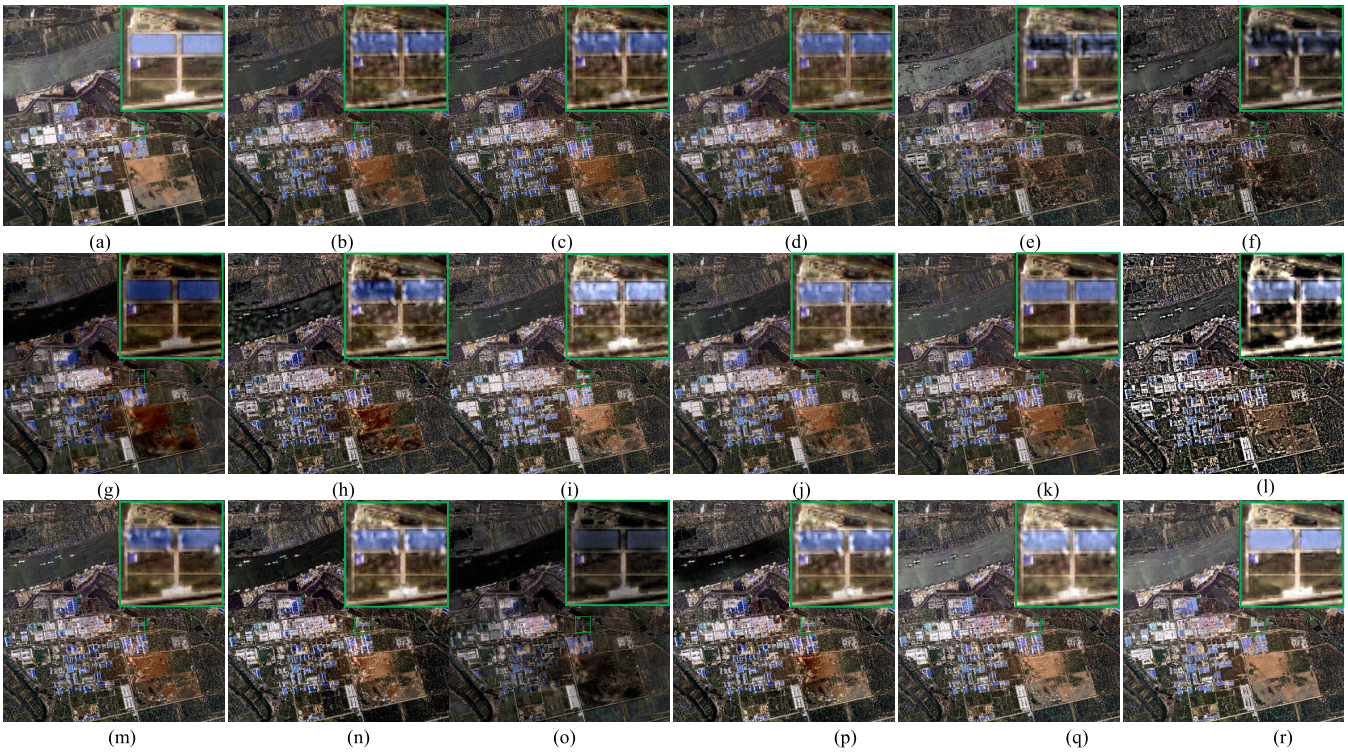


Fig. 8. Fusion results of different methods in area 2. (a) Original multispectral image, (b)–(q) the fusion images of the 16 compared methods, and (r) the fusion image of the propose method.

MDSDGF is high, but its spatial quality is poor. Of the five other methods, the spatial preservation ability and spectral fidelity are average, except for MS-NSST-PCNN, which retains spectral information better. Comparatively speaking, the designed method can clearly reflect detail information, such as road edge, building contour and river shoreline, and accurately express the spectral characteristics of various objects.

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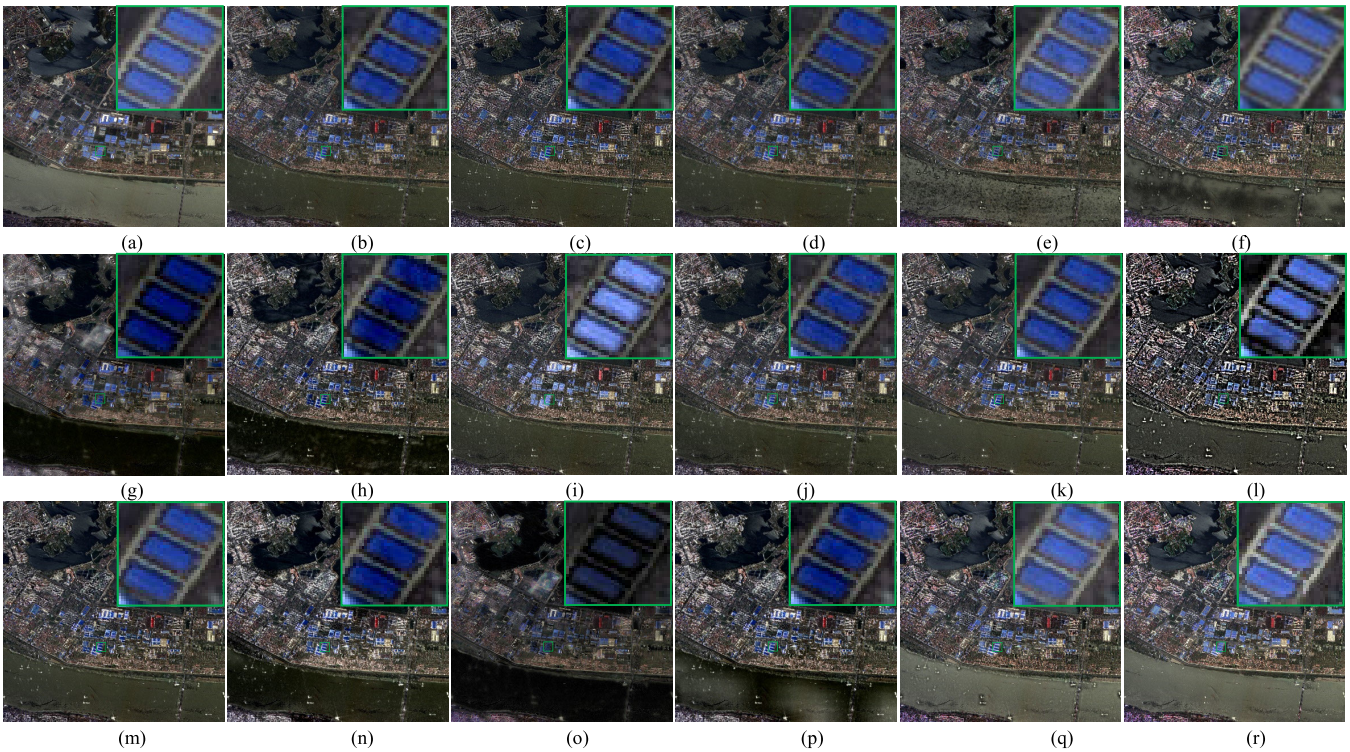


Fig. 9. Fusion results of different methods in area 3. (a) Original multispectral image, (b)–(q) the fusion images of the 16 compared methods, and (r) the fusion image of the proposed method.

As can be seen from the analysis of Fig. 9, the three MST methods do not perform well on the details of road edges, house contours, and the junction of different ground objects. Some ground object details and spectral information are lost. For example, the spectral characteristics of water bodies and vegetation are obviously different compared with the original images. Each of the three single-scale EPF methods has its disadvantages. JBF and RGF are not clear in spatial details and contour structure, and there are obvious artifacts. Although their spectral quality is better than that of the other comparison methods, there are still some shortcomings, such as the obvious speckle phenomenon in the water bodies. The spatial and spectral quality of GTF is worse than that of the other comparison methods. Compared with the other two kinds of methods, the five multiscale decomposition methods have obvious enhancement in spatial clarity, but there is still no significant improvement in spectral fidelity. In particular, the SD of MSGBF and MDLatLRR is relatively obvious, and the spectral features of the buildings in the zoomed-in region differ from the original multispectral images. Among the five other methods, the distortion of the spectra of MS-NSST-PCNN is the least pronounced, the spatial quality of CSMCA is poorer, and the SD of CNN is higher. Comparatively speaking, the method in this article shows excellent results in spatial clarity and spectral fidelity, and retains abundant spatial details and spectral characteristics in the original image.

2) Quantitative Evaluation: Quantitative evaluation is mainly based on information amount, spatial retention, spectral fidelity, and the degree of image distortion. Tables II–IV lists the results of all the methods in three sets of data, where the optimal value is in bold and the suboptimal value is in underline.

Table II shows the evaluation results in Area 1. In terms of IE and MI, two indexes for information amount, the three methods with the best effect are the proposed method, CNN, and MS-NSST-PCNN, successively. Compared with CNN, which ranks second, the proposed method has an increase of 0.037 and 0.073 in the two indexes, respectively, which indicates that the designed method has obvious superiority in preserving original image information. In terms of AG and SF, two indexes to evaluate the spatial quality of images, MSGBF, MSGRGF, and MDLatLRR have better results. However, compared with the designed method, there are still differences of 0.088 and 2.864 in the two evaluation indexes, indicating that the method has the highest spatial resolution. SD, ERGAS, and SAM are indexes to evaluate the degree of SD. The designed method, JBF, and MS-NSST-PCNN have obvious advantages over other methods, which reflects their excellent spectral fidelity. But comparatively speaking, the former has a slight advantage over the latter. On the QNR, which is a combined measure of image spatial and spectral quality, the proposed method improves 0.01 over the suboptimal value. For the evaluation index PSNR, only the proposed method, JBF, MDSDGF, and MS-NSST-PCNN exceed 8, but the former is slightly higher than the latter three, indicating that the image distortion degree of the proposed method is minimal.

Table III shows the results in Area 2. In terms of IE and MI, the designed method performs the best. In comparison with MSGRGF and MS-NSST-PCNN, which rank second and third, the fusion results are improved by 0.118 and 0.043,

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TABLE II
QUANTITATIVE EVALUATION RESULTS OF FUSED IMAGES IN AREA 1

Method	IE ↑	MI ↑	AG ↑	SF ↑	SD ↑	ERGAS ↓	SAM ↓	QNR ↑	PSNR ↑
Curvelet [40]	6.790	13.580	13.963	27.071	24.376	13.601	2.796	0.408	6.522
DTCWT [40]	6.884	13.768	14.311	27.856	26.681	14.931	3.290	0.367	5.899
NSCT [40]	6.698	13.396	11.837	23.497	23.153	12.998	2.522	0.412	6.786
JBF [34]	6.910	13.820	12.760	24.950	16.147	10.302	2.376	0.588	8.745
RGF [36]	6.977	13.953	13.456	26.638	29.655	17.380	3.343	0.406	4.770
GTF [41]	6.752	13.503	11.367	24.303	33.595	23.990	8.456	0.587	2.406
MSGBF [42]	7.084	14.168	15.828	31.423	34.269	20.429	6.554	0.274	3.849
MSGRGF [43]	7.029	14.060	16.305	31.323	25.242	13.243	2.626	0.363	7.064
MSGCF [44]	6.978	13.957	15.177	29.742	24.570	13.758	3.019	0.404	6.832
MDSDGF [45]	6.788	13.577	11.891	25.101	20.015	11.438	2.387	0.578	8.020
MDLatLRR [46]	6.862	13.729	16.240	31.366	34.895	18.977	5.394	0.447	5.246
CSMCA [47]	6.980	13.960	13.031	26.169	24.445	13.559	3.078	0.454	6.779
CNN [48]	7.190	14.383	14.965	29.127	31.240	17.325	5.479	0.356	5.311
IHS-GTF [17]	6.685	13.370	11.401	24.386	33.895	27.337	8.074	0.658	1.378
LP-SR [12]	7.127	14.258	15.076	29.200	43.755	21.942	3.815	0.366	2.813
MS-NSST-PCNN [49]	7.144	14.290	14.415	28.453	18.119	10.785	2.405	0.480	8.344
proposed	**7.227**	**14.456**	**16.393**	**34.287**	**15.856**	**10.218**	**2.084**	**0.598**	**8.996**

TABLE III
QUANTITATIVE EVALUATION RESULTS OF FUSED IMAGES IN AREA 2

Method	IE ↑	MI ↑	AG ↑	SF ↑	SD ↑	ERGAS ↓	SAM ↓	QNR ↑	PSNR ↑
Curvelet [40]	7.103	14.207	16.315	30.421	30.094	13.389	2.715	0.449	6.775
DTCWT [40]	7.182	14.366	16.665	31.052	32.813	14.551	3.230	0.397	6.191
NSCT [40]	7.009	14.018	13.708	26.394	28.197	12.712	2.324	0.468	7.114
JBF [34]	7.224	14.447	14.343	26.576	26.285	11.425	2.170	0.550	7.875
RGF [36]	7.086	14.171	14.783	27.712	40.606	21.000	2.840	0.368	3.033
GTF [41]	6.831	13.662	11.903	26.182	43.190	27.806	8.271	0.475	1.415
MSGBF [42]	7.291	14.583	18.140	34.845	39.317	19.399	5.335	0.369	4.132
MSGRGF [43]	7.352	14.106	**18.531**	34.806	28.707	11.658	2.377	0.446	8.198
MSGCF [44]	7.255	14.510	17.281	32.544	29.423	13.111	2.701	0.478	7.246
MDSDGF [45]	7.043	14.087	13.173	27.511	22.357	9.751	1.822	0.609	9.431
MDLatLRR [46]	7.125	14.257	16.978	31.128	42.029	17.903	5.530	0.476	5.459
CSMCA [47]	7.216	14.434	14.980	28.606	30.249	13.413	2.881	0.462	6.860
CNN [48]	7.302	14.607	17.001	32.048	39.660	19.704	5.963	0.419	4.483
IHS-GTF [17]	6.758	13.516	11.722	25.593	45.890	32.030	5.804	0.746	1.110
LP-SR [12]	7.337	14.877	16.437	32.177	47.368	20.743	4.118	0.395	3.409
MS-NSST-PCNN [49]	7.349	14.902	16.136	30.371	18.968	7.826	2.032	0.510	11.207
proposed	**7.470**	**14.945**	17.092	**34.870**	**13.774**	**6.818**	**1.672**	**0.685**	**12.357**

indicating that the designed method is able to integrate more image information. For the evaluation indexes AG and SF, the effect of the designed method, MSGBF, and MSGRGF is significantly superior to other methods. In AG, the designed method is inferior to MSGBF and MSGRGF, with poor spatial clarity. However, in SF, it is superior, indicating a high degree of spatial activity. Its results on SD, ERAGS, and SAM are 13.774, 6.818, and 1.672, respectively. Compared with the second-best values of 18.968, 7.826, and 1.822 on MDSDGF and MS-NSST-PCNN, it is optimized by 27.4%, 12.9%, and 8.2%, respectively, suggesting that the method in this article is able to suppress SD effectively. The method has obvious advantages in the evaluation index QNR and PSNR, which is optimized by 0.076 and 1.150 compared with the second-best value, which highlights its ability in image fidelity.

Table IV shows the results in Area 3, which are similar to those in Area 1 and Area 2. With reference to the results in Table IV, three MST methods are not dominant in all evaluation indexes, among which NSCT is better than curvelet and DTCWT in SD, ERGAS, and SAM, with better spectral quality, but worse than curvelet and DTCWT in AG and SF, with poorer spatial quality. The fusion quality of the three single-scale EPF methods is quite different. Among them, the three methods perform poorly on IE, MI, AG, and SF, with poor performance on spatial details. JBF is suboptimal on SD, ERGAS, and PSNR and optimal on QNR,

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TABLE IV
QUANTITATIVE EVALUATION RESULTS OF FUSED IMAGES IN AREA 3

Method	IE ↑	MI ↑	AG ↑	SF ↑	SD ↑	ERGAS ↓	SAM ↓	QNR ↑	PSNR ↑
Curvelet [40]	6.837	13.673	14.045	27.960	28.053	14.034	3.032	0.456	6.204
DTCWT [40]	6.923	13.846	14.400	28.748	30.826	15.562	3.475	0.402	5.496
NSCT [40]	6.748	13.495	11.838	24.154	26.552	13.319	2.793	0.469	6.530
JBF [34]	6.956	13.912	12.974	25.538	19.923	10.697	2.333	0.607	8.318
RGF [36]	7.093	14.186	13.651	27.936	31.102	14.678	1.847	0.446	5.790
GTF [41]	6.874	13.748	11.178	24.826	40.433	26.020	10.890	0.539	2.029
MSGBF [42]	7.162	14.324	15.802	32.516	42.699	24.337	8.602	0.300	2.657
MSGRGF [43]	7.049	14.099	**16.185**	31.749	30.043	14.648	2.769	0.403	6.112
MSGCF [44]	7.018	14.037	15.451	31.008	30.187	15.505	3.282	0.427	5.705
MDSDGF [45]	6.826	13.654	11.981	25.430	24.423	12.569	2.312	0.537	7.136
MDLatLRR [46]	6.961	13.927	15.135	32.490	40.148	20.368	5.593	0.464	4.528
CSMCA [47]	6.978	13.957	12.836	26.392	29.169	14.827	3.313	0.463	5.941
CNN [48]	7.184	14.370	15.060	30.022	41.705	21.249	6.820	0.339	3.654
IHS-GTF [17]	6.684	13.681	11.529	25.639	39.802	28.487	8.669	**0.699**	1.296
LP-SR [12]	7.208	14.237	15.067	30.010	39.688	20.282	6.538	0.353	3.884
MS-NSST-PCNN [49]	7.215	14.431	14.723	30.031	20.387	11.447	2.116	0.491	8.292
proposed	**7.254**	**14.510**	15.742	**33.544**	**17.645**	**10.508**	**1.736**	0.598	**8.571**

indicating better spectral fidelity and image fidelity. Five multiscale decomposition fusion methods have better fusion effect.

For AG and SF, except MDSDGF, the other four methods are superior to other comparison methods. In SD, ERGAS, SAM, QNR, and PSNR, MSGBF shows poor performance and serious SD. For the five other methods, three methods, CNN, LP-SR, and MS-NSST-PCNN, show better results in IE and MI, and all methods are unsatisfactory in AG and SF. On SD, ERGAS, and SAM, MS-NSST-PCNN shows a clear advantage in processing, second only to the proposed methods. his-GTF shows excellent results on QNR, and MS-NSST-PCNN has better results on PSNR. The designed method only performs slightly worse in the evaluation indexes AG and QNR, and is optimal in all other indexes, showing its obvious advantages. In comparison to the suboptimal values of all indexes, the designed method optimized 0.039, 0.079, 1.028, 2.278, 0.189, 0.111, and 0.253 for IE, MI, SF, SD, ERGAS, SAM, and PSNR, respectively, indicating that it is rich in information, has high spatial resolution, and spectral resolution, and is similar to the original image in structure.

3) Land Cover Classification Analysis: In this article, the land cover types of the three areas include road, building, woodland, water, and bare ground, represented by red, yellow, dark green, cyan, and dark brown, respectively. The random forest (RF) classifier is selected to classify the land cover of the original multispectral images, the fused images of the 16 compared methods and the proposed method, and the classification results are evaluated based on overall accuracy (OA) and kappa coefficient (KC). To ensure the rigor of the experiments, the same training samples and experimental environments are selected for all fused images for land cover classification [52]. The land cover classification results are shown in Fig. 10, and the classification accuracy is shown in Table V.

Since the experimental results of the three datasets are similar, Area 3 is taken as a representative for the land subcover class analysis. According to the classification results in Fig. 10, it can be seen that the original multispectral image can better show the distribution of various land cover types, thanks to its rich spectral information. The proposed methods and various fusion methods can combine the spectral information of multispectral images and the spatial information of SAR images, but all of these methods do not retain the spectral and spatial information to the same degree. Among them, the classification results of JBF, RGF, and MDLatLRR are more chaotic, with fragmentation of each land cover type. GTF, MSGBF, CNN, and IHS-GTF have poor retention ability of spectral information, which leads to serious confusion of land cover types in their classification results, especially the poor ability to distinguish between buildings and roads. The fusion results of several other methods incorporate the feature information of SAR images compared with the original multispectral images, and improve the differentiation of land cover types. However, due to the irrational design of the fusion rules, the distortion of the spectral information is exacerbated to different degrees, which reduces the interpretability of the fused images. The proposed method improves the above phenomena, the classification results are less affected by speckle noise, the degree of SD is not obvious, and different land cover types are distinguished more accurately.

Analyzed from Table V, MDLatLRR, RGF, CNN, and MSGBF have a serious loss of spectral information, which is reflected by the evaluation indexes SD, ERGAS, and SAM based on spectral information in Table I. These four methods also show poor performance in spatial resolution, so they are lower than the original multispectral image by 84.379% and 0.783 in both OA and KC evaluation indexes. Although JBF maintains better spectral characteristics, the poorer clarity leads to a lower classification accuracy than the

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TABLE V
LAND COVER CLASSIFICATION ACCURACY OF THE MULTISPECTRAL, PROPOSED METHOD, AND ITS FIVE VARIANTS IN AREA 1

Method	multispectral	curvelet	DTCWT	NSCT	JBF	RGF	GTF	MSGBF	MSGRGF
OA	84.379%	85.598%	84.922%	86.957%	82.381%	80.222%	85.385%	84.254%	85.253%
KC	0.783	0.796	0.786	0.815	0.753	0.722	0.791	0.775	0.791
Method	MSGCF	MDSDGF	MDLatLRR	CSMCA	CNN	IHS-GTF	LP-SR	MS-NSST-PCNN	proposed
OA	86.450%	86.854%	76.542%	84.819%	84.041%	87.383%	84.423%	87.662%	**90.306%**
KC	0.807	0.814	0.663	0.785	0.771	0.821	0.778	0.828	**0.865**

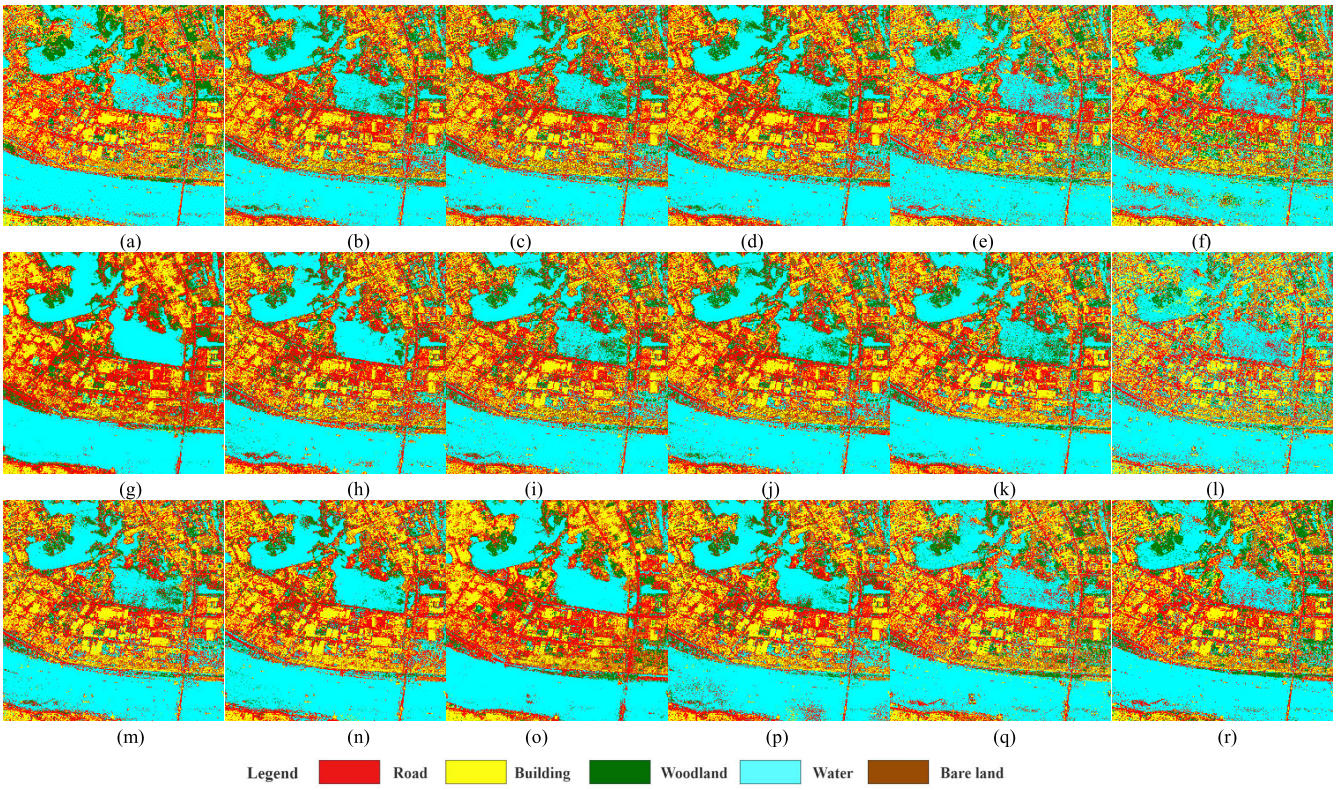


Fig. 10. Land cover classification results of different methods in area 3. (a) Original multispectral image, (b)–(q) the fusion images of the 16 compared methods, and (r) the land cover classification result of the proposed method.

original multispectral image as well. The spectral fidelity of LP-SR, CSMCA, DTCWT, MSGRGF, GTF, and curvelet is not outstanding and ranks in the middle of all the methods, which limits the effectiveness of land cover classification to some extent, especially the confusion between water, woodland, and bare land is more obvious. Their overall accuracy is distributed between 84% and 86%. MSGCF, MDSDGF, NSCT, IHS-GTF, and MS-NSST-PCNN improve the spatial resolution while retaining the spectral information better. Their classification accuracy has a large advantage over the other 11 compared methods, especially in the OA improvement is more than 2%, and the KC also has more than 0.02 improvement performance. The proposed method has strong spatial enhancement and spectral fidelity, which greatly improves the land cover classification accuracy. The OA and KC of its classification results are improved by 5.927% and 0.082 compared with the original multispectral image, and by 2.644% and 0.037 compared with the best of 16 other methods.

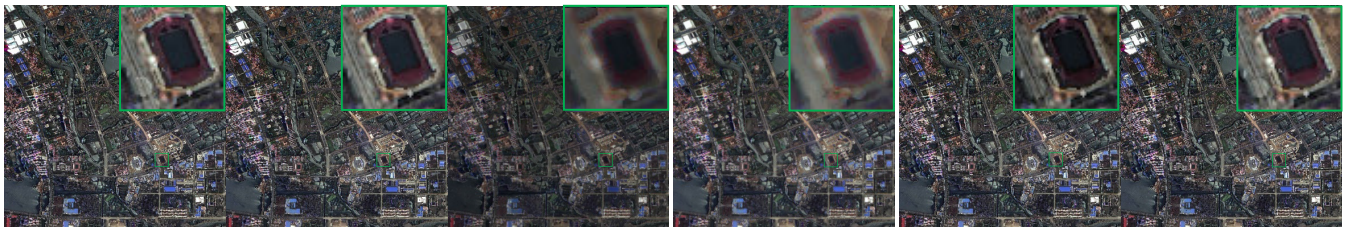
4) Fusion Performance Analysis: Five variants of the proposed method are designed separately for comparison with it to illustrate the effectiveness of the design of each part. The image decomposition structure of variant 1 is multiscale RGF (MSDRGF), and that of variant 2 is least squares optimized rolling filtering (LSRGF), and the multilevel fusion parts of both are consistent with the proposed method. The large-scale layer fusion method for variant 3 and 4 is absolute value maximization, and the base layer fusion method for variant 3 and 5 is weighted average i.e., $B_F(i,j) = 0.5B_{\mathrm{SAR}}(i,j) + 0.5B_{\mathrm{MS}}(i,j)$. The image decomposition structure and other layer fusion methods for all three remain consistent with the proposed method. The design details of the proposed method and the five variants are represented in TABLE VI. In addition, all fused images are classified for land cover and analyzed for application performance.

The fusion results are analyzed according to the proposed method and its five variants are in Figs. 11–13. Variant 1 is

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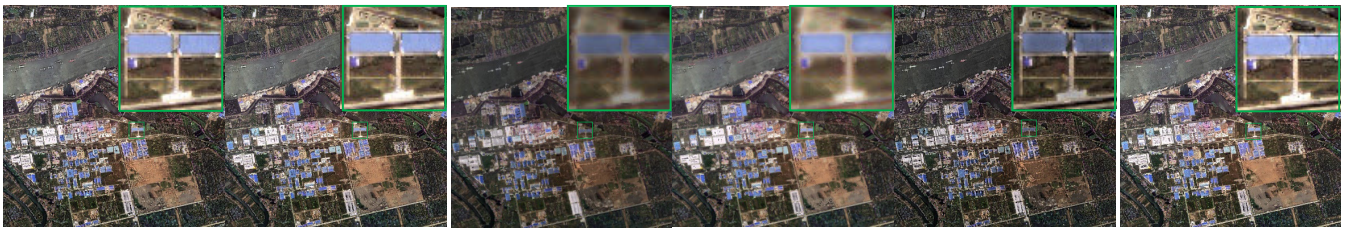
TABLE VI
DESIGN DETAILS OF THE PROPOSED METHOD AND ITS FIVE VARIANTS

Methods	Image decomposition structure	Multi-level fusion		
		Small-scale layer	Large-scale layer	Base layer
variant1	MSDRGF	absolute value maximization	CNP system	WLE、WSEML
variant2	LSRGF	absolute value maximization	—	WLE、WSEML
variant3	MSDLSRGF	absolute value maximization	absolute value maximization	weighted average
variant4	MSDLSRGF	absolute value maximization	absolute value maximization	WLE、WSEML
variant5	MSDLSRGF	absolute value maximization	CNP system	weighted average
proposed	MSDLSRGF	absolute value maximization	CNP system	WLE、WSEML



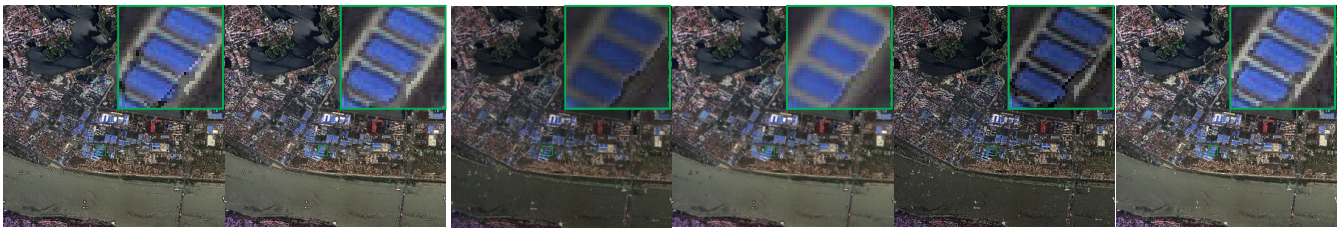
(a) (b) (c) (d) (e) (f)

Fig. 11. Fusion results of the proposed method and its 5 variants in area 1. (a) Variant1. (b) Variant2. (c) Variant3. (d) Variant4. (e) Variant5. (f) Proposed.



(a) (b) (c) (d) (e) (f)

Fig. 12. Fusion results of the proposed method and its 5 variants in area 2. (a) Variant1. (b) Variant2. (c) Variant3. (d) Variant4. (e) Variant5. (f) Proposed.



(a) (b) (c) (d) (e) (f)

Fig. 13. Fusion results of the proposed method and its 5 variants in area 3. (a) Variant1. (b) Variant2. (c) Variant3. (d) Variant4. (e) Variant5. (f) Proposed.

not optimized by least squares. Its spatial resolution and spectral fidelity are slightly inferior, reflecting that the least-squares optimized design can enhance the overall quality of the fused image. Variant 2 is not multiscale decomposed compared to the proposed method. Its spatial blurring is more obvious and the spectral quality is reduced, indicating that the multiscale decomposition operation is able to decompose more image information to a greater extent. Variant 3, variant 4, and variant 5 are respectively adjusted compared to the strategy of the designed method. Among them, the fusion strategies of the large-scale and base layers of variant 3 are designed as absolute value maximization and weighted average, respectively. As shown in the zoomed-in region, its spatial clarity appears to be significantly degraded, and the color is dark and the SD is obvious. Variant 4 has an improved base layer compared to variant 3, and its spectral features have been optimized to be very close to the original multispectral image, but its spatial resolution has not been significantly improved. Variant 5 has improved the large-scale layer compared with variant 3, and its spatial blurring phenomenon has been significantly improved, but its SD phenomenon is still relatively serious. The proposed method

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TABLE VII
QUANTITATIVE EVALUATION RESULTS OF THE PROPOSED METHOD AND ITS FIVE VARIANTS IN AREA 1

Method	IE ↑	MI ↑	AG ↑	SF ↑	SD ↑	ERGAS ↓	SAM ↓	QNR ↑	PSNR ↑
variant1	7.188	14.377	14.782	30.307	17.392	10.330	2.135	0.553	8.907
variant2	7.171	14.344	14.165	29.342	16.901	10.622	2.150	0.583	8.493
variant3	6.492	12.985	5.519	12.490	24.686	14.294	2.592	0.540	5.727
variant4	6.893	13.788	6.466	14.673	15.963	**8.345**	**1.892**	**0.667**	**10.372**
variant5	6.910	13.820	14.113	28.943	26.897	15.757	3.289	0.420	5.528
proposed	**7.227**	**14.456**	**16.393**	**34.287**	**15.856**	10.218	2.084	0.598	8.996

TABLE VIII
QUANTITATIVE EVALUATION RESULTS OF THE PROPOSED METHOD AND ITS FIVE VARIANTS IN AREA 2

Method	IE ↑	MI ↑	AG ↑	SF ↑	SD ↑	ERGAS ↓	SAM ↓	QNR ↑	PSNR ↑
variant1	7.462	14.929	15.796	31.690	17.012	8.054	1.726	0.601	10.919
variant2	7.449	14.902	15.111	30.352	15.102	7.205	1.686	0.635	11.865
variant3	6.769	13.539	4.491	11.087	28.571	13.739	2.104	0.645	6.159
variant4	7.251	14.503	5.644	13.180	14.760	**6.097**	**1.539**	**0.778**	**13.092**
variant5	7.146	14.293	15.348	30.823	32.779	15.378	2.597	0.459	5.656
proposed	**7.470**	**14.945**	**17.092**	**34.870**	**13.774**	6.818	1.672	0.685	12.357

TABLE IX
QUANTITATIVE EVALUATION RESULTS OF THE PROPOSED METHOD AND ITS FIVE VARIANTS IN AREA 3

Method	IE ↑	MI ↑	AG ↑	SF ↑	SD ↑	ERGAS ↓	SAM ↓	QNR ↑	PSNR ↑
variant1	7.246	14.493	14.339	30.293	19.833	10.998	1.910	0.548	8.184
variant2	7.220	14.441	13.826	29.422	18.412	10.525	1.778	0.571	8.408
variant3	6.544	13.088	5.210	12.979	26.442	14.036	2.458	0.579	5.817
variant4	6.949	13.898	6.536	15.718	**16.798**	**8.666**	**1.699**	**0.654**	**9.880**
variant5	6.905	13.810	13.512	28.539	30.498	16.328	2.939	0.439	5.132
proposed	**7.254**	**14.510**	**15.742**	**33.544**	17.645	10.508	1.736	0.598	8.571

improves the large-scale and base layers over variant 3 with higher spatial resolution and spectral fidelity, indicating that the design of large-scale layer fusion based on the CNP system and the base layer fusion design combining WLE and WSEML have greater advantages in enhancing the spatial and spectral information.

According to Tables VII–IX, the quantitative evaluation results of the three areas are similar, so Area 1 is used as an example for analysis. Variant 1 was designed to verify the effectiveness of least squares optimization. According to Table VII, it suggests the designed method is optimized by 0.039, 0.079, 1.611, 3.980, 1.536, 0.112, 0.051, 0.045, and 0.089 in nine evaluation indexes compared to variant 1, indicating that least squares optimization of RGF can effectively suppress artifacts such as gradient inversion and halos, respectively, thereby enhancing the extraction of spatial and spectral information.

Variant 2 is designed to prove the effect of the multiscale decomposition design, and the designed method optimizes 0.056, 0.112, 2.228, 4.945, 1.045, 0.404, 0.066, 0.015, and 0.503 over variant 2 in all evaluation indexes, respectively, indicating that extending the RGF to multiscale decomposition can achieve the scale decomposition of space overlapping features, thus facilitating the image information fusion. Variant 3, variant 4, and variant 5 are designed for multilevel fusion strategies, where variant 3 changes the fusion strategies of large-scale and base layers and replaces them with two simple strategies of absolute value maximization and weighted average, respectively. The fusion results show that variant 3 becomes significantly worse in all evaluation indexes, which indicates that the unreasonable design of fusion rules can seriously affect the quality of fused images. Variants 4 and 5 were designed to prove the effect of the CNP system fusion strategy and the fusion strategy combining WLE and WSEML. Compared with variant 3, the optimization of variant 4 is very obvious in five indexes, SD, ERGAS, SAM, QNR, and PSNR, by 8.723, 5.949 0.700, 0.127, and 4.645, respectively, indicating that the base layer fusion design combining WLE and WSEML can greatly enhance the spectral fidelity of the images. The designed method has significant improvements in IE, MI, AG, and SF compared with variant 4, indicating that the large-scale layer fusion design with the CNP system can effectively integrate image information, especially detail information. But there is a deterioration in four indexes,

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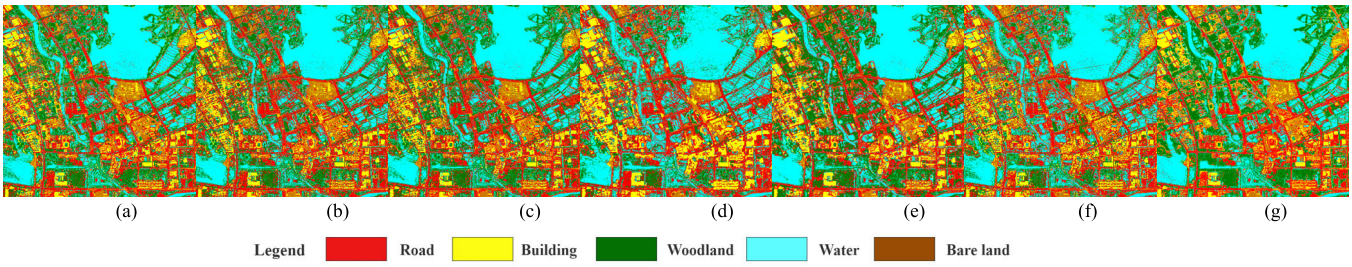


Fig. 14. Results of the multispectral, proposed method and its five variants for land cover classification in area 1. (a) Multispectral. (b) Variant1. (c) Variant2. (d) Variant3. (e) Variant4. (f) Variant5. (g) Proposed.

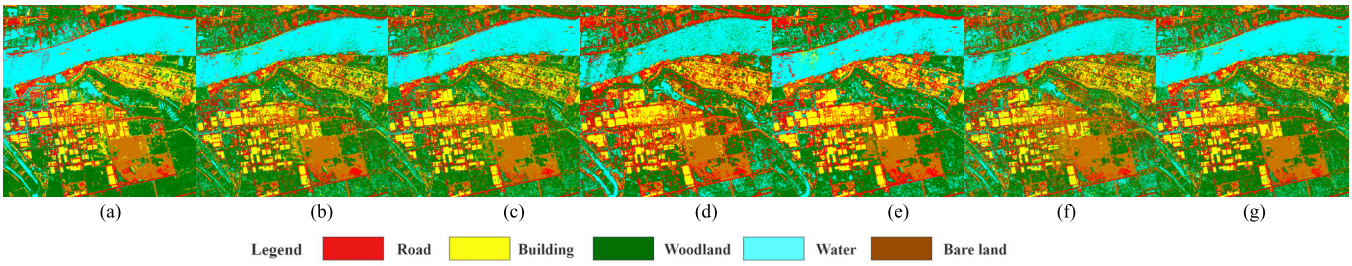


Fig. 15. Results of the multispectral, proposed method and its five variants for land cover classification in area 2. (a) Multispectral. (b) Variant1. (c) Variant2. (d) Variant3. (e) Variant4. (f) Variant5. (g) Proposed.

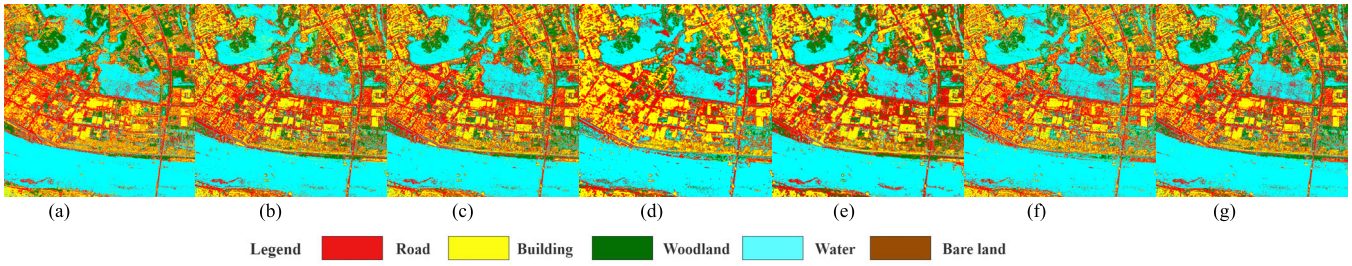


Fig. 16. Results of the multispectral, proposed method and its five variants for land cover classification in area 3. (a) Multispectral. (b) Variant1. (c) Variant2. (d) Variant3. (e) Variant4. (f) Variant5. (g) Proposed.

ERGAS, SAM, QNR, and PSNR, indicating that the CNP system can affect the spectral quality of the image to some extent when the spatial information is greatly enhanced, which is a common problem in most fusion methods and needs further research to solve. However, this fusion method, which sacrifices a small amount of spectral quality and largely improves the spatial quality to enhance the overall capability of the fusion method, is also a feasible strategy. Variant 5 has a great degree of improvement over variant 3 in two indexes for evaluating the spatial quality of images, AG and SF, by 8.594 and 16.453, respectively, indicating that the large-scale layer fusion design with the CNP system can effectively enhance the spatial quality of images. Compared with variant 5, the proposed method is optimized in all evaluation indexes, with the most obvious optimization in five indexes, SD, ERGAS, SAM, QNR, and PSNR, reflecting the outstanding ability of the base layer fusion design combining WLE and WSEML in terms of spectral fidelity. In addition, the fusion strategy combining WLE and WSEML did not degrade the spatial quality while substantially improving the image spectral fidelity, indicating that the fusion strategy combining WLE and WSEML is capable of extracting detail information while maintaining energy. It also improves the spectral fidelity and sharpness of the images.

According to Figs. 14–16, it can be seen that the original multispectral images can produce good classification results based on the rich spectral information, which can better show the distribution of various land cover types. Compared with the original multispectral images, both the proposed method and its variants combine the feature information of SAR to improve the differentiation between landcover types, but the unreasonable design of the fusion rules aggravates the distortion of the spectral information, reduces the interpretability of the fused images, and largely affects the overall classification results. This is greatly improved in the proposed method, the classification results are less affected by speckle noise, the degree of SD is not obvious, and there are finer classification results than the original multispectral images in the details and edges of the classification, and the land cover types are distinguished more accurately.

As shown in Table X, the SD of variant 1, variant 2, and variant 3 is serious, and the improvement of spatial resolution is not prominent, which leads to their overall classification accuracies being lower than that of the original multispectral images. The overall classification accuracies of variant 3 and variant 4 are slightly better than that of the original multispectral image, and the improvements in OA and KC are no more than 1% and 0.02, respectively. The

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TABLE X
LAND COVER CLASSIFICATION ACCURACY OF THE MULTISPECTRAL, PROPOSED METHOD, AND ITS FIVE VARIANTS IN AREA 1

Method	multispectral	variant1	Variant2	Variant3	Variant4	Variant5	proposed
OA	76.898%	75.062%	76.496%	77.456%	77.843%	75.836%	**80.928%**
KC	0.685	0.661	0.680	0.689	0.699	0.668	**0.738**

TABLE XI
LAND COVER CLASSIFICATION ACCURACY OF THE MULTISPECTRAL, PROPOSED METHOD, AND ITS FIVE VARIANTS IN AREA 2

Method	multispectral	variant1	Variant2	Variant3	Variant4	Variant5	proposed
OA	85.668%	82.278%	85.118%	77.159%	83.563%	72.588%	**89.436%**
KC	0.812	0.766	0.803	0.700	0.784	0.642	**0.860**

TABLE XII
LAND COVER CLASSIFICATION ACCURACY OF THE MULTISPECTRAL, PROPOSED METHOD, AND ITS FIVE VARIANTS IN AREA 3

Method	multispectral	variant1	Variant2	Variant3	Variant4	Variant5	proposed
OA	84.379%	86.112%	88.793%	86.905%	87.485%	85.187%	**90.306%**
KC	0.783	0.808	0.844	0.815	0.827	0.791	**0.865**

proposed method shows better results in terms of the amount of information, clarity, and spectral information, and the OA and KC of the classification results have been improved by 4.03% and 0.053, respectively, compared to the original multispectral image, and by 3.0% and 0.053, respectively, compared to the best variant of five variants. The OA and KC of the classification results are improved by 4.03% and 0.053, respectively, compared with the original multispectral images, and 3.085% and 0.039, respectively, compared with the best of the five variants.

According to the analysis performed in Table XI, all the variants showed severe SDs, and none of the classification accuracies were as good as the original multispectral image, however, the proposed method was optimized over the original multispectral image by 3.768 and 0.048 in OA and KC, respectively. Table XII shows that all variants have significantly enhanced the differentiation of the detail part compared with the original multispectral image, and variant 1 and variant 5 are still affected by SD, and the improvement of classification accuracy compared with the original image is less than 2% and 0.03 in both OA and KC evaluation indexes. Variant 2, variant 3, and variant 4 can better extract the detail information and spectral information in the original image, and thus have a better ability to improve the classification accuracy of the land cover. The proposed method has better spatial information retention and spectral information fidelity, and the OA and KC of the classification results are improved by 5.927% and 0.082 compared with the original multispectral image, and 1.513% and 0.021 compared with the method with the best classification effect among the five variants, respectively.

The proposed method, like the vast majority of traditional methods, does not fully consider the heterogeneity between heterogenic images, such as the differences in contour structure and gray-scale texture. For this reason, in the future, the feature differences of heterogenic images will be considered in depth to design more reasonable fusion methods. In addition, the proposed method does not retain ship outline information well, and some ship detail information is lost, which is very unfavorable for projects such as shipping traffic monitoring. For this reason, combining the advantages of SAR and multispectral imagery in ship identification is considered in the future to design ship monitoring methods that are beneficial to shipping management.

IV. CONCLUSION

In the article, a multispectral and SAR image fusion method for multiscale decomposition based on LS optimization RGF is proposed. Based on LS, RGF is optimized, all gradients in the image are constrained, and artifact phenomena such as gradient reversals and halos are solved. By combining the optimized RGF filtering and GF smoothing speckle noise in the image, the detail information, edge information, and basic information in the image are decomposed simultaneously. The image decomposition is further extended to multiscale space to achieve scale separation of space overlapping features and retain more details. Then, corresponding to the decomposition degree, all the information falls into three layers. According to characteristics of different scale layers, fusion rules are designed according to the absolute value maximization, CNP system, WLE, and WSEML, respectively, thereby enhancing the extract ability of detail information and improving the fusion quality. The proposed method is compared with 16 methods, among which nine representative indexes are adopted for evaluation. The results show that, in comparison with the optimal values of all comparison methods, the proposed method is obviously optimized on all indexes. It suggests that the proposed method has better retention ability in spatial and spectral information.

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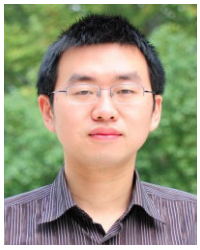
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